

Review Questions for the Midterm Exam

1 Policies and Overview

- midterm exam policies
- overview of problems, algorithms, data structures
- overview of discrete mathematics

2 Sample Questions

- on the cost functions of algorithms
- the Gale-Shapley algorithm
- algorithms on graphs
- greedy algorithms
- divide and conquer

CS 401/MCS 401 Lecture 10
Computer Algorithms I
Jan Verschelde, 10 July 2026

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midterm exam policies

- The Midterm Exam happens on Monday 13 July, from 10AM till 11:40AM.
- The exam covers the first 9 lectures.
This corresponds to the first five chapters in the textbook.
Sections not explicitly covered in class will not be on the exam.
- There will not be a makeup Midterm Exam.
- You may skip the Midterm Exam.
The 100 points transfer automatically to the Final exam.

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overview of problems

- 1 stable marriage (L-1, L-3)
- 2 graph traversals (L-3, L-4)
- 3 connected components in graph (L-4, L-5, L-8)
- 4 testing bipartiteness (L-4)
- 5 ordering nodes in a directed acyclic graph (L-5)
- 6 scheduling requests on one or multiple resources (L-5, L-6)
- 7 optimal caching (L-6)
- 8 shortest paths in a graph (L-6)
- 9 spanning trees (L-7)
- 10 counting inversions (L-8)
- 11 closest pair of points (L-8)
- 12 convolutions (L-8, L-9)
- 13 matrix multiplication (L-9)

overview of algorithms

- 1 Gale-Shapley algorithm (L-1, L-3)
- 2 breadth first search (L-3, L-4)
- 3 depth first search (L-3, L-4)
- 4 topological ordering (L-5)
- 5 interval scheduling (L-5)
- 6 interval partitioning (L-5)
- 7 earliest deadline first (L-6)
- 8 farthest in the future (L-6)
- 9 Dijkstra's algorithm (L-6)
- 10 Kruskal's algorithm (L-7)
- 11 Prim's algorithm (L-7)
- 12 reverse delete (L-7)
- 13 merge sort to count inversions (L-8)
- 14 closest pair of points (L-8)
- 15 Karatsuba's integer multiplication (L-8)
- 16 fast convolution (FFT) (L-9)
- 17 Strassen's method (L-9)

overview of data structures

- 1 array and list (L-3)
- 2 queue and stack (L-4)
- 3 adjacency matrix representation of a graph (L-4)
- 4 edge list representation of a graph (L-4, L-5)
- 5 priority queue or heap (L-7)
- 6 union-find data structure (L-7)

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overview of discrete mathematics

- 1 asymptotic bounds: big O , Ω , and Θ (L-2)
- 2 logarithmic, sublinear, linear, subquadratic, quadratic, cubic, polynomial, exponential growth (L-2)
- 3 transitivity and sum properties (L-2)
- 4 solving a recurrence by substitution (L-8)
- 5 solving a recurrence by unrolling (L-8)
- 6 the master method to solve recurrences (L-9)

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1. derive and prove a cost function

Consider the following problem:

Input: $U = [u_{i,j}]$, an n -by- n matrix, $u_{i,i} \neq 0$,

$\mathbf{b} = (b_1, b_2, \dots, b_n)$, a vector of length n .

Output: $\mathbf{x} = (x_1, x_2, \dots, x_n)$, a vector of length n , with

$$x_i = \frac{1}{u_{i,i}} \left(b_i - \sum_{j=i+1}^n u_{i,j} x_j \right), \quad i = n, n-1, \dots, 1.$$

- 1 Express the cost to compute $\mathbf{x}$ as a function of n .
- 2 Justify your cost function.

2. compare in the O , Ω , Θ order

Let $f(n) = n\sqrt{100n}$ and $g(n) = n^2$.

- 1 Consider the statements:

$f(n)$ is $O(g(n))$, $f(n)$ is $\Omega(g(n))$, $f(n)$ is $\Theta(g(n))$.

Which statement is true?

For each statement, justify your answer.

- 2 Consider the statements:

$g(n)$ is $O(f(n))$, $g(n)$ is $\Omega(f(n))$, $g(n)$ is $\Theta(f(n))$.

Which statement is true?

For each statement, justify your answer.

3. solving a recurrence

Apply the master method to solve the following recurrences:

- 1 $T(n) = 9T(n/3) + n$
- 2 $T(n) = T(n/2) + 1$
- 3 $T(n) = 2T(n/2) + n \log(n)$

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4. the Gale-Shapley algorithm

Consider 4 employers $\{e_1, e_2, e_3, e_4\}$ and 4 applicants $\{a_1, a_2, a_3, a_4\}$.

preferences for employers

e_1 : 1 3 2 4

e_2 : 2 1 4 3

e_3 : 1 3 2 4

e_4 : 4 1 3 2

preferences for applicants

a_1 : 2 3 4 1

a_2 : 1 4 2 3

a_3 : 1 3 4 2

a_4 : 2 1 3 4

Preferences are listed from high to low for employers and applicants.

- 1 Run the Gale-Shapley algorithm on the above input.
Show all stages of the algorithm.
- 2 To demonstrate that the algorithm is efficient, a data structure is needed to determine the next applicant an employers offers to.
What is this data structure?
Illustrate the use of this data structure on the above input.

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5. topological ordering of a directed acyclic graph

Consider the topological ordering in a DAG.

- 1 Let G be a graph with 6 vertices 0, 1, 2, 3, 4, 5 and edges $E = \{ (0, 1), (0, 2), (0, 5), (1, 2), (3, 1), (2, 4), (2, 5), (3, 4), (4, 5) \}$. Illustrate the algorithm to order the vertices topologically on the above example.
Draw all steps in the execution of the algorithm.
- 2 Is the solution of a topological ordering unique?
Justify your answer.

6. relabeling nodes in a tree

In a tree, assume that the n nodes are labeled

- starting with label 0 at the root, and
- ending at label $n - 1$ of the last visited node in an inorder traversal of the nodes.

Answer the following questions:

- 1 Given any positive label $k < n$, is it possible to reorganize the nodes of the tree so that the node with label k becomes the root?
- 2 Define an algorithm and the data structure used to store the tree.
- 3 Use big O to express the cost of your algorithm.

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7. shortest paths in a graph

Consider the shortest paths in a graph.

- 1 Let G be a weighted graph with 6 vertices 0, 1, 2, 3, 4, 5 with lengths ℓ on edges (i, j) written as $e_{(i,j)} = \ell$ below:
 $e_{(0,1)} = 7$, $e_{(0,2)} = 9$, $e_{(0,5)} = 14$, $e_{(1,2)} = 10$, $e_{(1,3)} = 15$,
 $e_{(2,3)} = 11$, $e_{(2,5)} = 2$, $e_{(3,4)} = 6$, $e_{(4,5)} = 9$.

Compute the shortest paths originating at 0.

Draw all steps in the execution of the algorithm.

- 2 Use the step wise execution of the algorithm above to illustrate the main argument in the correctness proof of the algorithm.

In particular, if the set of explored vertices equals three, justify why the path to the fourth vertex in the path from 0 will be the shortest path from 0 to the fourth explored vertex.

8. minimum spanning tree

- 1 Let G be a weighted graph with 7 vertices 0, 1, 2, 3, 4, 5, 6 with weights w on edges (i, j) written as $e_{(i,j)} = w$ below:
 $e_{(0,1)} = 4$, $e_{(0,3)} = 1$, $e_{(1,2)} = 10$, $e_{(1,3)} = 7$, $e_{(1,4)} = 5$, $e_{(2,4)} = 2$,
 $e_{(3,4)} = 9$, $e_{(3,5)} = 3$, $e_{(4,5)} = 8$, $e_{(4,6)} = 6$, $e_{(5,6)} = 11$.

Execute Kruskal's algorithm on the above example.
Draw all steps in the execution of the algorithm.

- 2 Illustrate the evolution of the union-find data structure during the execution of Kruskal's algorithm on the above example.
- List the main arguments why using the union-find data structure leads to an $O(m \log(n))$ running time for Kruskal's algorithm, for a graph with n vertices and m edges.

9. interval scheduling

Let the tuples $(1, 8)$, $(2, 14)$, $(10, 16)$, $(3, 19)$, $(17, 20)$, and $(18, 21)$ represent the start and finish times of 6 requests for a resource.

- 1 Run the scheduling algorithm to minimize maximum lateness on these requests.
Show all steps in the execution of the algorithm.
- 2 What is the exchange argument used to show optimality?
Illustrate the exchange argument on an example.

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10. counting inversions

Consider the sequence 8, 5, 2, 4, 6, 1, 3, 7.

- 1 Illustrate the algorithm to count the number of inversions on the above sequence.
- 2 Explain why the counting does not lead to more operations than the number of operations one needs for merge sort.

Illustrate the key argument by referring to the illustration on the above sequence.

11. divide and conquer

For a number x and positive integer n ,
the direct computation of x^n requires $n - 1$ multiplications.

- 1 Explain how divide and conquer leads to an algorithm which requires $O(\log_2(n))$ multiplications.
- 2 Formulate the algorithm and prove the bound on its running time.